EDGE OLYMPICS | PARTICIPANT GUIDE

30 October 2026 · Submission deadline: **6:00 PM India Standard Time (IST)** · Official portal: edgeolympics.vercel.app

Build a single TensorFlow Lite image classifier and submit it through your team page. The organizers validate the file, then benchmark eligible models on Raspberry Pi 4B workers against a private evaluation set.

1 Get started

- Sign in on the event site. Create a team or join one with the team's invite code. Submit from **My team**.
- Use the training data and class definitions supplied for the challenge. Keep a separate validation split while developing. The official evaluation images and labels are private.
- Train on your own computer or cloud compute if useful. Only the exported model runs during judging; no notebook, script, network service, or sidecar file accompanies it.

2 Exact model contract

File	One self-contained **.tflite** file, at most **50 MiB** (50 × 1,048,576 bytes).
Input	One image tensor of shape **[1, 128, 128, 1]**: batch 1, grayscale. Input image values represent the normalized **0–1** range.
Input dtype	**float32, int8, or uint8**. For quantized inputs, the judge uses the model's input scale and zero point.
Output	First output tensor has shape **[1, 8]** and contains eight class scores. The highest score determines the predicted class, indexed **0–7** in the supplied class order.
Runtime	The model must load and run with the official CPU TFLite interpreter on a Raspberry Pi 4B. Avoid custom or delegate-only operators.

Check the exact input/output tensors and run a local inference before uploading. A correct extension alone does not mean the model will pass validation.

3 Submit and track

- Open **My team**, select the .tflite file, and press **Submit model** before 6:00 PM IST. The portal displays the deadline in your device's local timezone.
- Each team has **three submission slots**. Pending, running, completed, and disqualified entries use slots. A failed entry releases its slot so you can fix the model and submit again.
- Follow the result on **My team** or **Leaderboard / My submissions**. Normal progress: pending_prelim, queued_for_edge, running, completed. A failed status includes an error to help diagnose the issue.
- There is no overwrite or 'latest file wins' behavior. Each upload creates a separate submission; the best completed eligible score for your team is used for ranking.

4 How scoring works

The benchmark measures **accuracy** on the hidden set, **inference latency** in milliseconds, and **peak RAM** in MiB. The current weights are accuracy 60%, latency 25%, and RAM 15%, with reference values of 100 ms and 256 MiB. The organizer dashboard can change settings; the live leaderboard shows the active values.

Score = 100 × [0.60 × accuracy + 0.25 × 100/(100 + latency_ms) + 0.15 × 256/(256 + peak_ram_mb)]. Accuracy is a fraction from 0 to 1. Higher score is better. Ties use accuracy (higher), then latency (lower), then RAM (lower).

Model file size is a submission limit and displayed metric, but it is not part of the weighted score. Temperature is monitored for safe operation; it is not a scored metric. There is no minimum accuracy qualification threshold.

5 Before you upload

- Confirm the file is self-contained, under 50 MiB, and has the required shape, dtype, normalized image domain, and class order.
- Run it locally with a CPU TFLite interpreter and check that all eight output scores are finite. Leave time for validation and, if necessary, another upload before the deadline.
- Do not attempt to access the hidden evaluation set, another team's files, or judging infrastructure. Ask an organizer about event logistics or a submission status you cannot resolve.

Portal: edgeolympics.vercel.app | This guide describes the current portal and benchmark contract.